Automobiles and curb along rue de Terrebonne

Registered vehicles known by postal district, modelled below it; legal curb measured block face by block face

Rick Hoge, PhD November 12, 2026 version 2026-11-12-01-06-45

DRAFT for the author's review, November 12, 2026. Not yet checked line by line; figures may change.

Lay summary

How many cars belong to the households along the Terrebonne bike lane, and how much curb is there to park them? Québec's vehicle registry, as released, places each vehicle only in a postal district of about 10,000 households (H4A or H4B), not on a street. This note looks closer, and says which parts are measured and which estimated.

Measured: the curb. Using the City's inventory of parking signs, every side of every block was read for stretches where parking is always forbidden. The 22 census areas that touch the lane, reaching about one block to each side, hold 37 km of legal curb. That is an upper limit: driveways and hydrants are not removed.

Estimated: the cars. Census facts about each small area (renters, commuters who drive, household size, income) were used to share out each district's registered automobiles. For the corridor's 5,471 households the estimate is about 4,600 automobiles, between 4,400 and 4,900: 0.84 per household. For a single small area the estimate is uncertain by about a quarter either way.

Tested where the answer is known (sharing a group of districts' total among its members), the method was off by 5 per cent at the median, against 14 per cent for a split by number of households.

At 5.5 m per car, the corridor's legal curb would hold about 6,800 cars: more than the estimated number registered there, even if all were kept on the street. Every lot was examined on aerial photos and a random sample of 112 checked from the street: 85 of 99 small buildings have a driveway, pad or garage. That makes about 2,200 private places, garage interiors and apartment buildings not counted. Neither figure measures parking difficulty: visitors, time limits and distance to one's door are left out.

The note makes no recommendation.

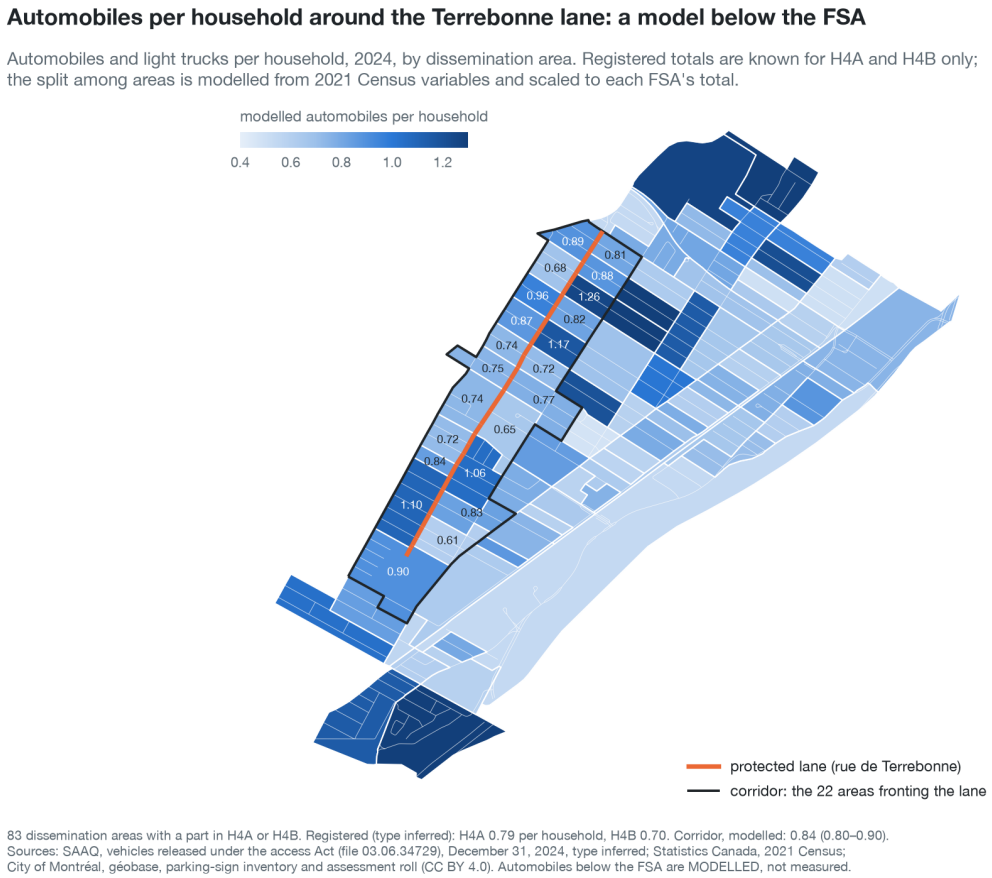


Figure 1: Overview. Modelled automobiles and light trucks per household by dissemination area, 2024. The registered totals are known for the two FSAs only; the split among areas is a model.

Contents

1 Question **4**

2 Data **4**

3 Method **4**

3.1 Dissemination areas and FSAs 4

3.2 The corridor 5

3.3 Legal curb 5

3.4 Automobiles below the FSA 5

4 Results **6**

4.1 Does the model work where the answer is known? 6

4.2 The corridor 7

4.3 Legal curb 8

4.4 Automobiles beside curb: if every vehicle were kept on the street 8

4.5 Private parking 9

4.6 A check from the 2015 LiDAR night 12

5 What the note shows and does not show **12**

6 Reproducibility **13**

A Tables **13**

1 Question

The vehicle files the SAAQ released under the access Act for 2023 and 2024 (file 03.06.34729) locate each vehicle to its forward sortation area (FSA, the first three characters of the postal code), and no finer. The protected bike lane on rue de Terrebonne lies in two FSAs: H4B west of avenue Royal, H4A east of it. On December 31, 2024 they held 7,351 and 9,598 automobiles and light trucks (type inferred from make, model and model year, as in the note of the repository *Quebec-vehicle-registrations*), that is 0.70 and 0.79 per household of the 2021 Census.

An FSA holds ten to twelve thousand households; the lane's surroundings are a fraction of each. This note asks what can be said at a finer grain, about the dissemination areas that front the lane: how many automobiles are registered there, and how much legal curb they hold. The two halves of the answer are of different kinds, and the note keeps them apart throughout:

- **Curb is measured**, block face by block face, from the City's géobase and its inventory of parking signs.
- **Automobiles below the FSA are modelled**, from 2021 Census variables, and scaled so that the areas of each FSA sum to its registered total. Within an FSA the split is an estimate with an interval, never a count.

The 2015 LiDAR and the aerial imagery read for the 1954–55 plans work serve as a check, not as an input. The note makes no recommendation.

2 Data

- **Vehicles.** SAAQ, files released under the access Act for December 31, 2023 and 2024, by FSA; the table `type_by_fsa.csv` of *Quebec-vehicle-registrations* (type inferred; all automobiles and light trucks, whether registered to a person or to a business: the release carries no class).
- **Census.** Statistics Canada, 2021 Census profile by FSA (98-401-X2021013) and by dissemination area (98-401-X2021006), and the dissemination-area boundaries. Counts are randomly rounded to 5 by Statistics Canada; tenure, commuting and income come from the 25 % sample.
- **Curb.** City of Montréal, géobase (street segments, October 14, 2026) and *Signalisation – stationnement* (sign inventory, October 23, 2026), both CC BY 4.0.
- **Dwellings.** City of Montréal, *Unités d'évaluation foncière* (assessment roll, CC BY 4.0), used to share a dissemination area that straddles two FSAs and to count the dwellings addressed on rue de Terrebonne; FSA polygons from DMTI Spatial, CanMap Postal Code Suite v2024.3 (under licence, not redistributed).
- **Checks.** The cars counted by eye on the City's 2015 night LiDAR and the garage and driveway readings, from the folder *1954-underwriters-plans-terrebonne-garages*; the parking counts of the borough's 2023 study (EXP), as transcribed in *2026-parking-supply-demand-survey-context*.

3 Method

3.1 Dissemination areas and FSAs

Dissemination areas do not nest in FSAs. Each area is shared among FSAs in proportion to the dwellings of the assessment roll found in each FSA polygon. Rebuilt this way from dissemination areas,

the households of the island's 96 FSAs of 1,000 households or more differ from Statistics Canada's own FSA profile by 0.5 % at the median (H4A −1.2 %, H4B −1.2 %). 83 areas have a part in H4A or H4B.

3.2 The corridor

The corridor is the set of 22 dissemination areas that hold a block face of rue de Terrebonne along the lane, from its west end to Girouard. They reach about one block to each side (Somerled is about 260 m north of the lane, Monkland about 285 m south); 17 of them have nine-tenths or more of their surface within 300 m of the lane (the others are the larger areas toward the west end, the smallest share being 19 %), and no area outside the set has half its surface within that distance.

3.3 Legal curb

A block face is one side of one géobase segment (autoroutes, bridges, tunnels and back lanes left out). A segment runs from the centre of one intersection to the centre of the next. A face's legal curb is its length, less 10 m at each end, less the stretches where a sign forbids stopping or parking at all times (« \A EN TOUT TEMPS », « \P EN TOUT TEMPS »). The stretch a sign governs runs from its pole to the next pole in the direction of its arrow; a sign without an arrow governs the stretches on both sides of its pole. Where the usual reading of the arrows leaves a stretch claimed both by an all-time prohibition and by another parking rule (hours, days, a time limit: a rule that implies parking exists there), the other reading is taken for that face if it leaves fewer such stretches; a stretch still claimed both ways counts for half.

The 10 m at each end is the half roadway of the cross street (5 m, an assumption: the local cross streets measure 9.0 to 9.4 m from curb to curb on the 2015 LiDAR, rue de Terrebonne 11.0 m) plus the 5 m next to an intersection where the Highway Safety Code forbids stopping, signed or not (art. 386, par. 4).

The result is an **upper bound** on legal curb: driveways, hydrants, bus stops, hourly rules, permit zones and seasonal rules are not deducted. Places are legal curb divided by 5, 5.5 and 6 m.

Check on rue de Terrebonne, where the layout is known: the reading finds parking on the south side west of Cavendish and on the north side east of it, one band, as built in 2024: 1,306 m of legal curb, 238 places at 5.5 m. The borough's 2023 study counted 478 places on the two sides before the lane (report p. 40): 230 westbound and 248 eastbound.

3.4 Automobiles below the FSA

1. Across the 114 FSAs beginning with H that hold 1,000 households or more, the logarithm of automobiles per household is regressed on four census variables: the share of renter households, the share of commuters who drive, persons per household, and the logarithm of the median household income. The regression is robust (Huber weights), because a few FSAs hold fleets registered to businesses (H2Y, H2Z, H3A, H3B, H4P, H9P: registered counts more than 1.5 times what their census variables predict).
2. The rate of each dissemination area is predicted from its own census variables.
3. The predictions are scaled so that the parts of the areas lying in each FSA sum to that FSA's registered total. The scale factors of H4A and H4B are 1.07 and 0.99.

Specifications were compared by leaving each FSA out in turn (Table 4); the one used has the smallest error at the 90th percentile and close to the smallest at the median. Dwelling type adds nothing once tenure, commuting and household size are in.

Intervals come from 2,000 draws, each combining: the FSAs resampled and the model refitted; the sampling error of the 25 % sample variables in each dissemination area; an unexplained factor for each area, whose spread is the one measured by the downscaling test below; and the scaling to the FSA totals. Dissemination areas are about twenty times smaller than FSAs, and their true unexplained spread may be larger than the one measured among FSAs: the tables also carry intervals with that spread doubled, which are wider by a few points only, because the sampling error of the census variables dominates. The intervals do not cover what the model cannot see: the vehicles are those of December 31, 2024 and the census is of May 2021; the type is inferred; vehicles registered to businesses are in the totals.

4 Results

4.1 Does the model work where the answer is known?

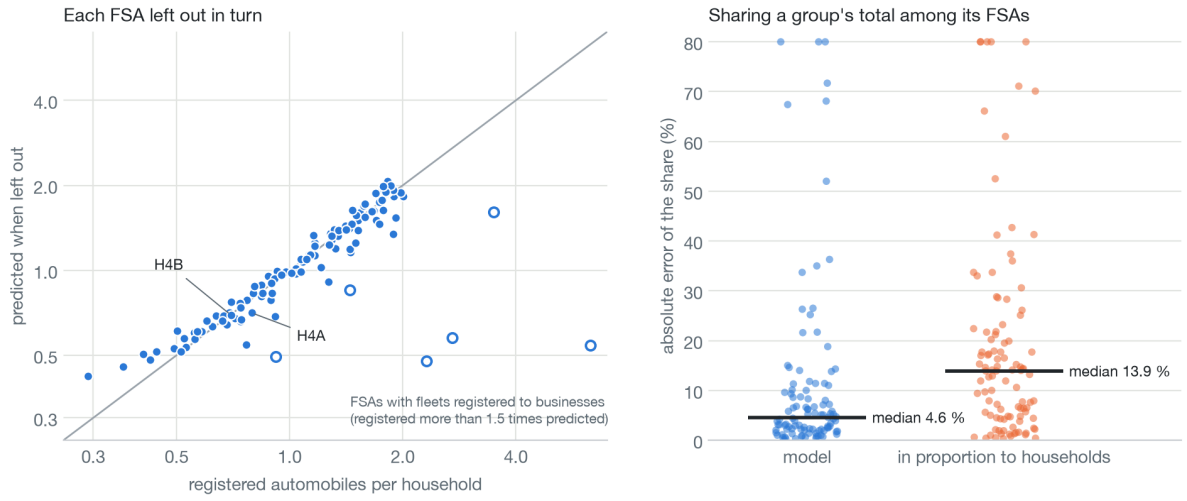
Two tests, both at the level of FSAs, the only level at which registered counts exist.

Each FSA left out in turn. The model fitted on the other FSAs predicts the left-out FSA's rate within 5.4 % at the median (24 % at the 90th percentile, the business FSAs included). H4A is under-predicted by 11 %, H4B by 0.6 %.

Downscaling, one level up. Neighbouring FSAs were grouped by threes (27 groups, 103 FSAs); for each group the model was fitted without its members, and the group's registered total was shared among them by the model. This is the operation the note performs below the FSA, done where it can be checked. The share of each FSA is off by 4.6 % at the median (26 % at the 90th percentile); sharing in proportion to households is off by 13.9 % (42 %). Without the groups that hold a business FSA: 3.9 % against 12.3 %. For H4A and H4B taken as one group, the model's shares are off by −4.6 % and +6.0 %, the household split by −5.5 % and +7.1 %: for this pair the model gains little.

Testing the model where the answer is known: FSAs

Left: each FSA's rate predicted by a model fitted without it. Right: a group of neighbouring FSAs' registered total shared among its members, by the model and in proportion to households; error of each member's share.



Sources: SAAQ, vehicles released under the access Act (file 03.06.34729), December 31, 2024, type inferred; Statistics Canada, 2021 Census; City of Montréal, géobase, parking-sign inventory and assessment roll (CC BY 4.0). Automobiles below the FSA are MODELLED, not measured.

Figure 2: The two tests. Left: each FSA's rate predicted without it (hollow: FSAs with fleets). Right: error of each FSA's share of its group's total; errors above 80 % drawn at 80.

The tests say the model shares a total among FSAs better than households alone do. They cannot say how well it shares an FSA among its dissemination areas: no registered count exists at that level.

4.2 The corridor

Table 1: Modelled automobiles and measured legal curb. Interval: 90 %. Places: legal curb at 6 m to 5 m per vehicle (upper bound). Last column: modelled automobiles per place at 5.5 m.

	areas	hh.	autos	interval	per hh.	curb km	places	per place
Corridor	22	5,471	4,622	4,382–4,927	0.84	37.2	6,192–7,431	0.68
Other areas of H4A, H4B	61	18,978	14,363	14,033–14,641	0.76	109.3	18,210–21,852	0.72
All	83	24,449	18,985	18,830–19,163	0.78	146.4	24,402–29,283	0.71

The corridor's 5,471 households are estimated to have 4,622 automobiles and light trucks registered (4,382 to 4,927), 0.84 per household (0.80 to 0.90); the other areas of the two FSAs, 0.76. The two alternative specifications give 4,616 and 4,683 for the corridor. Area by area the estimates run from 0.61 to 1.26 per household, and the 90 % interval of a single area is about 29 % of its estimate either way (Figure 3, Table 6). The differences between neighbouring areas are mostly within those intervals.

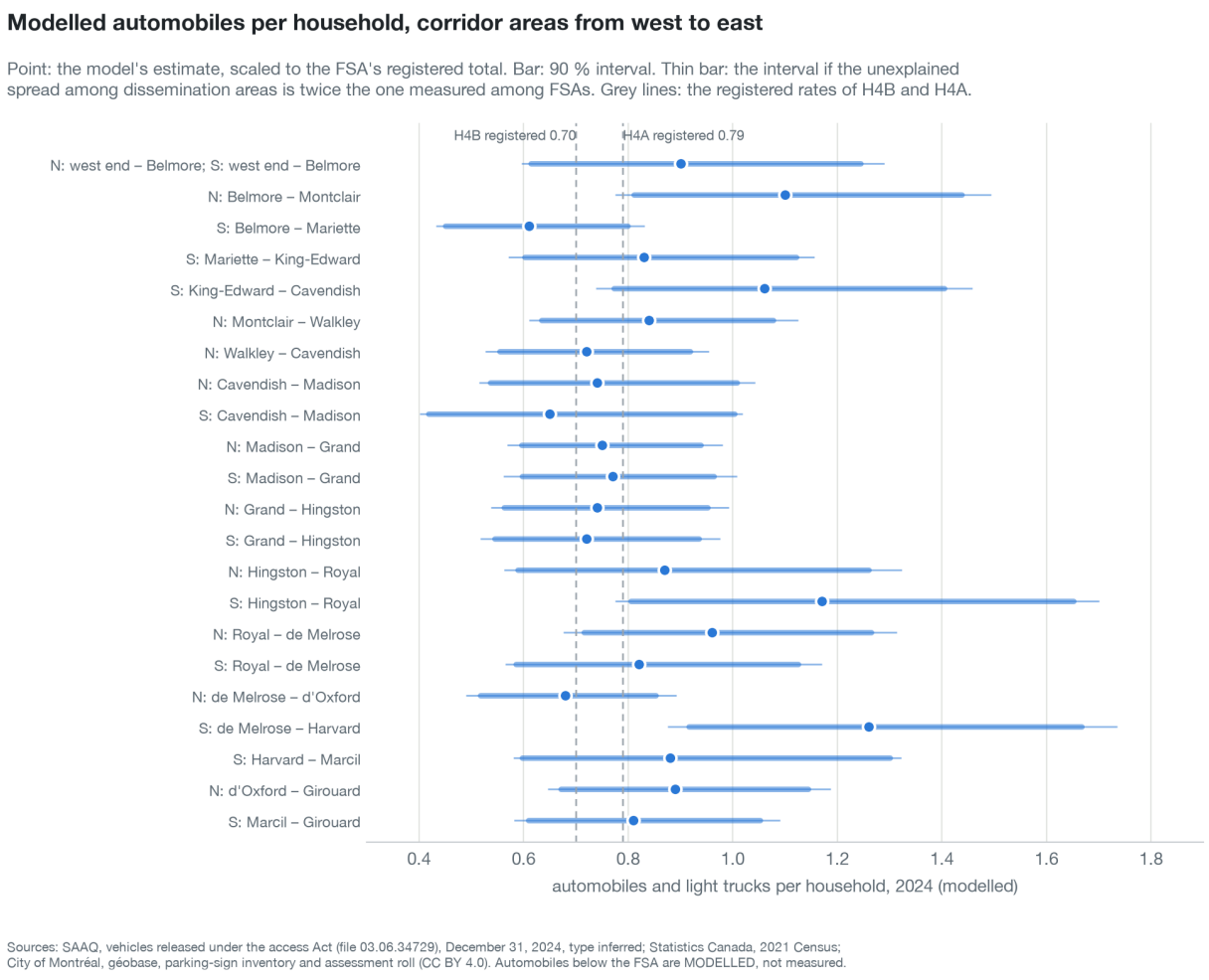


Figure 3: Modelled automobiles per household, the corridor's areas from west to east.

4.3 Legal curb

The corridor holds 46.8 km of curb; net of 10 m at each end of each face, 8 % is forbidden at all times, leaving 37.2 km of legal curb: 6,192 to 7,431 places at 6 to 5 m, an upper bound. That is 6.8 m of legal curb per household (5.8 m in the other areas of the two FSAs).

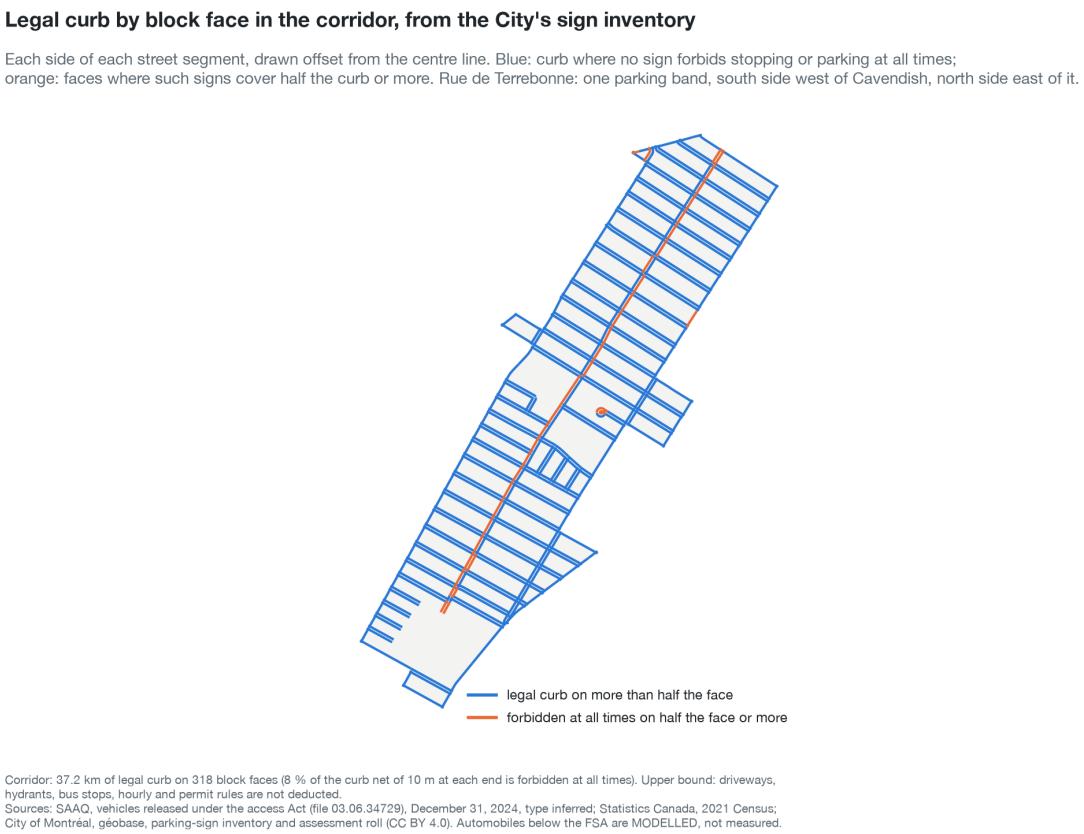


Figure 4: Block faces of the corridor. A face is drawn orange when all-time prohibitions cover half of it or more.

4.4 Automobiles beside curb: if every vehicle were kept on the street

At 5.5 m per place the corridor has 0.68 modelled automobile per place of legal curb (0.65 to 0.73); in 6 of the 22 areas the estimate exceeds the area's own places (Figure 5).

This comparison is the limiting case in which every registered vehicle is kept at the curb and no private place is used. Private parking is taken up in the next section.

The ratio is not a measure of parking pressure, in either direction. The curb figure is an upper bound (no driveway, hydrant or bus stop deducted, no hourly or permit rule). The automobile figure is every vehicle registered, wherever it is kept. Visitors, institutions and vehicles from elsewhere use the same curb; a place at the far end of an area is not a place at one's door. The measurements of occupancy that exist are those of the borough's 2023 study: on Thursday, September 29, 2022, at 18:30–19:30, 247 cars on rue de Terrebonne's places (50 %) and 311 on the cross-street stubs (51 %) (EXP, Tables 3-2 and 3-3, as transcribed in *2026-parking-supply-demand-survey-context*).

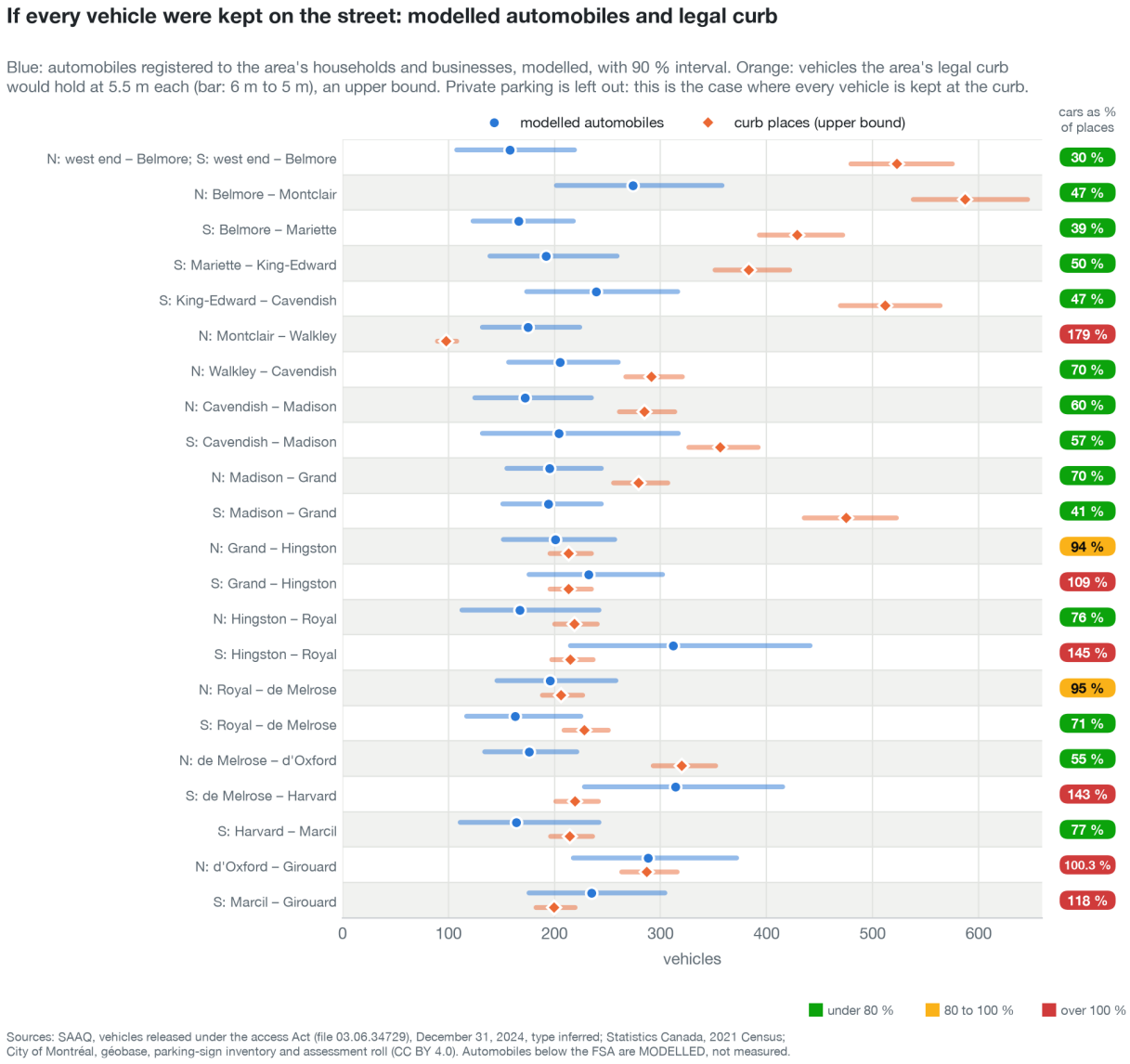


Figure 5: Modelled automobiles and the places the legal curb would hold, by area.

4.5 Private parking

Measured, on rue de Terrebonne. The buildings addressed on the street were read twice in the folder *1954-underwriters-plans-terrebonne-garages*: on the Underwriters' plans of 1954–55 (a garage as built) and on present-day aerial imagery (a driveway or paved apron). Of the 123 buildings read, 89 have one or the other (72 %; 95 % interval 64 to 80). Counting one private place for each such building gives 89 places for 268 dwellings, whose modelled automobiles number 226; the street's one parking band holds 238 places at 5.5 m. One place per building understates capacity where a driveway holds more than one car, and overstates it where a garage of 1954 is no longer used for a car. The larger apartment buildings and the frontages whose plan sheet is missing were not read.

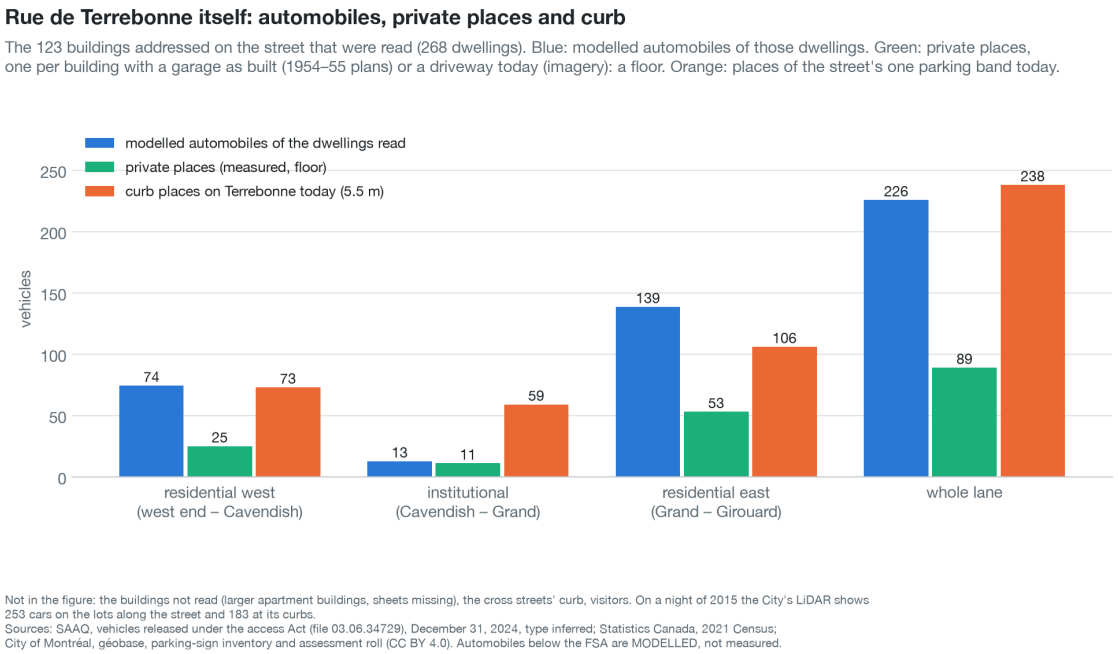


Figure 6: Rue de Terrebonne: modelled automobiles of the dwellings in the buildings read, private places (one per building with a garage or driveway), and the places of the street's parking band today, by sector of the 2023 study.

Measured, for the corridor. Every lot of the corridor that holds a dwelling (2,358 lots) was read on enlarged crops of the same aerial imagery, by independent readers working from a written protocol (folder `derived/readings/`): can a car be driven off the street and left standing clear of the sidewalk, on the lot's own driveway, a driveway shared with the neighbour, a front pad, a rear pad or a parking lot? A first reading failed its checks and was discarded; the protocol and the images were corrected and every lot was read again. Two independent readings of the same lot then agreed on 95 % of the lots where both gave a yes or a no.

Agreement between readers is not accuracy. Before the second reading, 112 lots were drawn at random (100 buildings of five dwellings or fewer, 12 larger), and the author judged each one from street-level imagery, without seeing any reader's answer. Of the 99 small buildings judged yes or no, 85 have car access (86 %; 95 % interval 78 to 91). The readers from above missed part of them:

Table 2: The author's check against the readers' answers, buildings of five dwellings or fewer. Places: as counted by the author; independent = a car can leave without another being moved; garage interiors not counted.

Reader's answer	lots checked	with car access	independent places	total places
Yes	56	52	59	97
Unclear	20	18	20	36
No	23	15	16	26

The estimate therefore rests on the check, and uses the readings only to sharpen it: to the lots of each reader answer (1,428 yes, 470 unclear, 364 no) it applies the share with access and the mean number of places the author found for the checked lots of that answer. Intervals come from resampling the checked lots (2,000 draws, 90 %).

For the corridor's 2,262 buildings of five dwellings or fewer: 1,986 have car access (1,867 to 2,096; 88 %), with 2,228 independent places (2,051 to 2,414) and 3,731 places in all when cars standing one behind another are counted (3,415 to 4,053). Garage interiors are in neither figure. Curb and independent private places together number 8,983 at 5.5 m, against 4,622 modelled automobiles: 0.51

automobile per place, against 0.68 with the curb alone; the estimate exceeds the area's places in 2 of the 22 areas, against 6.

Buildings of six dwellings or more (96 buildings, 2,592 dwellings) are given no private place here. Of the 12 in the check, 6 have an underground garage whose places cannot be seen from the street or from above, 4 have no parking, and for 2 it could not be told. Their modelled automobiles are in the comparison; their garages are not.

Limits. The check is one person's judgment from street-level imagery, on 99 small buildings. The aerial imagery predates the 2024 lane; the street-level imagery is more recent. The reading of rue de Terrebonne of October 25 above was made from the air alone and, like the readers here, will have missed part of the driveways: its 72 % is lower than the 86 % found for the corridor from the street.

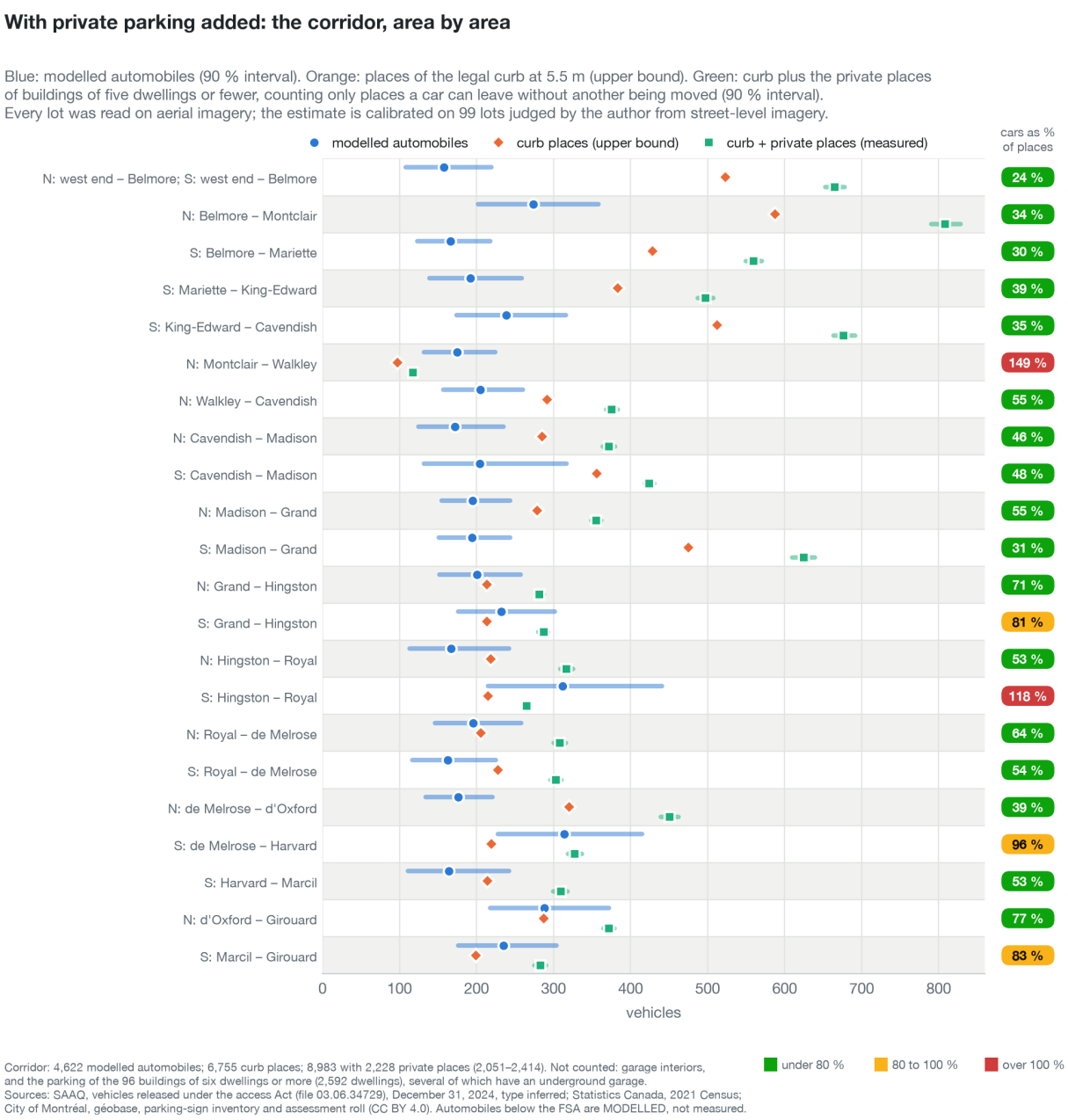


Figure 7: The corridor's areas with the measured private places of small buildings added (independent places only).

4.6 A check from the 2015 LiDAR night

The City's 2015 LiDAR was flown at night. Counted by eye along rue de Terrebonne: 183 cars at the curbs and 253 on the lots along the street, 436 for the 573 dwellings addressed on the street: 0.76 per dwelling. The model gives 0.84 per household for the areas that hold those dwellings.

Table 3: Cars seen on the 2015 LiDAR night along rue de Terrebonne, and the model, by sector of the 2023 study.

Sector	dwellings	at curbs	on lots	seen per dwelling	modelled per household
Residential west	146	52	97	1.02	0.93
Institutional	140	29	45	0.53	0.66
Residential east	287	102	111	0.74	0.89
Whole lane	573	183	253	0.76	0.84

The two quantities are not the same: the LiDAR sees visitors' cars and the lots of churches, schools and corner buildings addressed on the cross streets, and does not see cars inside garages or away for the night; nine years separate the two. They agree in order of magnitude. They do not agree in detail: across the 11 areas with ten dwellings or more on Terrebonne, the rank correlation between cars seen per dwelling and the modelled rate is 0.31 (p = 0.36), not distinguishable from none. The check therefore supports the level of the estimate, not the split between areas.

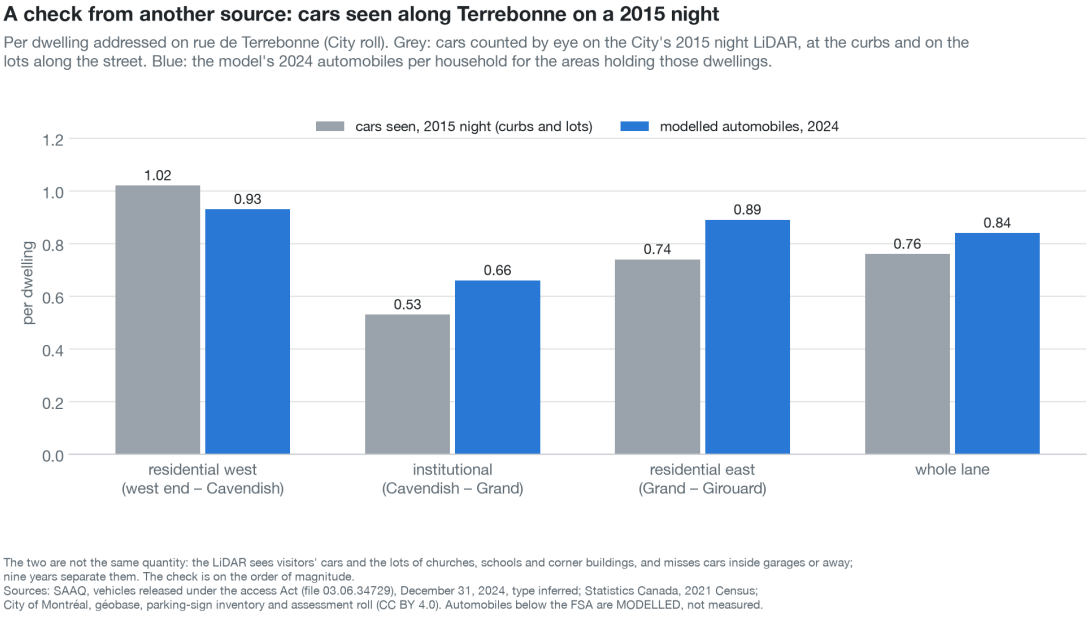


Figure 8: Cars seen per dwelling on the 2015 night and the model's rate, by sector.

5 What the note shows and does not show

- **Measured:** the registered automobiles of H4A and H4B (type inferred); the legal curb of each block face, as an upper bound; the dwellings and, on rue de Terrebonne, the garages and driveways.
- **Measured, with a measured error:** private places of the corridor's small buildings, from a reading of every lot calibrated on a random sample checked from the street.
- **Not counted:** garage interiors; the parking of buildings of six dwellings or more, several of which have underground garages.

- **Modelled:** the automobiles of every area smaller than an FSA. The model shares a known total better than households alone at the one level where this can be tested; below the FSA nothing can test it, and the 2015 check does not confirm its area-by-area differences.
- **Not measured by anyone:** where registered vehicles are kept; occupancy of the curb since the lane was built.
- The curb reading rests on the sign inventory as published on October 23, 2026 and on rules for arrows that were checked on rue de Terrebonne only.

6 Reproducibility

Scripts, tables and figures: `https://github.com/OpenMobilityData/Terrebonne-open-sources/tree/main/2026-terrebonne-corridor-vehicles-and-curb`. Order: `build_dataset.py`, `curb_supply.py`, `demand_model.py`, `corridor.py`, `offstreet_strips.py`, `offstreet_crops.py`, `offstreet_readings_v2.py`, `offstreet.py`, the figure scripts under `derived/figures/`, `build_docs.py`. The census, roll, sign and postal-code files are read from outside the repository (sources and dates in the folder's README); the tables by dissemination area and by block face are in `derived/tables/`.

A Tables

Table 4: Specifications compared, each FSA left out in turn: absolute error of the predicted rate (%), and the signed error for H4A and H4B.

Variables	median	75th	90th	H4A	H4B
households only	35.9	50.2	98.3	+40.0	+58.9
tenure, dwelling type	14.7	29.8	51.0	+29.9	+6.4
commuting	7.9	13.8	25.2	−15.3	+2.4
tenure, commuting, household size	5.5	10.9	25.8	−10.8	+0.2
tenure, commuting, household size, income (used)	5.4	11.0	24.4	−10.7	−0.6
tenure, commuting, household size, income, dwelling type	6.1	11.3	27.0	−9.9	−2.1

Table 5: Coefficients of the model used (logarithm of automobiles per household; robust regression, 114 FSAs).

Variable	coefficient	standard error
constant	−2.620	0.783
share of renter households	+0.015	0.114
share of commuters who drive	+2.013	0.114
persons per household	+0.198	0.047
log of median household income	+0.091	0.066

Table 6: The corridor's dissemination areas, west to east. N, S: side of rue de Terrebonne and the cross streets the frontage runs between. Automobiles: modelled, with 90 % interval. Places: legal curb at 5.5 m (upper bound).

Frontage on Terrebonne	FSA	hh.	autos	interval	per hh.	curb km	places
N: west end – Belmore; S: west end – Belmore	H4B	176	158	108–219	0.90	2.88	523
N: Belmore – Montclair	H4B	249	274	202–358	1.10	3.23	587
S: Belmore – Mariette	H4B	273	166	123–218	0.61	2.36	429

Table 6: The corridor's dissemination areas, west to east. N, S: side of rue de Terrebonne and the cross streets the frontage runs between. Automobiles: modelled, with 90 % interval. Places: legal curb at 5.5 m (upper bound).

Frontage on Terrebonne	FSA	hh.	autos	interval	per hh.	curb km	places
S: Mariette – King-Edward	H4B	231	192	139–259	0.83	2.11	384
S: King-Edward – Cavendish	H4B	225	239	174–316	1.06	2.82	512
N: Montclair – Walkley	H4B	208	175	132–224	0.84	0.54	98
N: Walkley – Cavendish	H4B	283	205	157–260	0.72	1.60	292
N: Cavendish – Madison	H4V	233	172	125–235	0.74	1.57	285
S: Cavendish – Madison	H4B	316	204	132–317	0.65	1.96	356
N: Madison – Grand	H4B	260	195	155–244	0.75	1.54	279
S: Madison – Grand	H4B	253	194	151–244	0.77	2.61	475
N: Grand – Hingston	H4B	270	201	152–257	0.74	1.17	213
S: Grand – Hingston	H4B	323	232	176–302	0.72	1.17	213
N: Hingston – Royal	H4A	192	167	113–242	0.87	1.20	219
S: Hingston – Royal	H4A	267	312	215–441	1.17	1.18	215
N: Royal – de Melrose	H4A	204	196	146–258	0.96	1.13	206
S: Royal – de Melrose	H4A	200	163	117–225	0.82	1.25	228
N: de Melrose – d'Oxford	H4A	259	176	134–221	0.68	1.76	320
S: de Melrose – Harvard	H4A	249	314	228–415	1.26	1.21	220
S: Harvard – Marcil	H4A	186	164	111–242	0.88	1.18	214
N: d'Oxford – Girouard	H4A	325	288	218–372	0.89	1.58	287
S: Marcil – Girouard	H4A	289	235	176–304	0.81	1.10	199